Real-Time Impact Point Prediction of Suborbital Rockets Using Kalman Filtering

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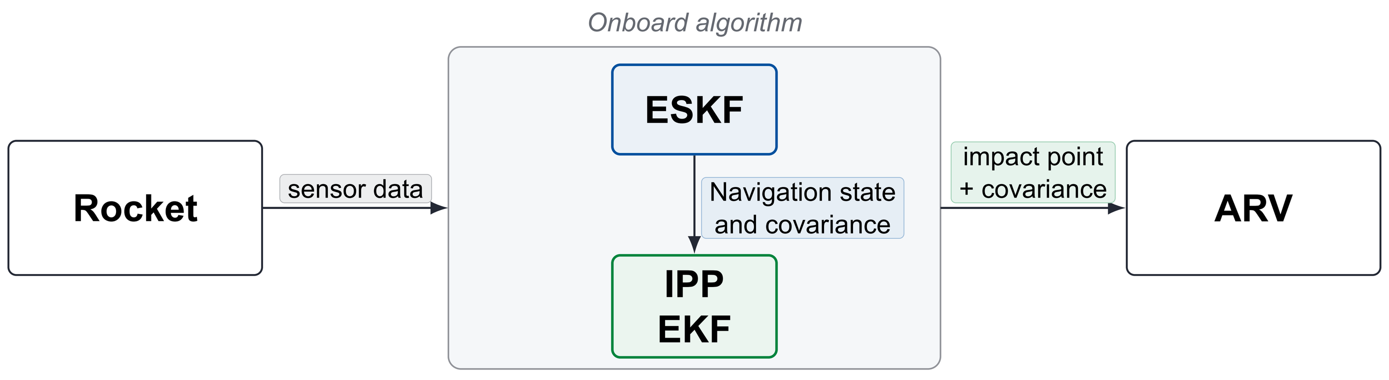
Motivation

Rocket launch rates are rising. Recovering payloads and first stages using autonomous recovery vehicles (ARVs) reduces cost, human exposure, and environmental footprint. The challenge is knowing where the rocket will land *before it gets there*.

Impact point prediction (IPP) is the real-time estimate of the rocket's landing location and uncertainty, updated continuously from onboard sensor data. An ARV steers to this estimate.



System & Method

Navigation — Error-State Kalman Filter (ESKF)

The ESKF maintains a continuous estimate of position, velocity, attitude, and sensor biases. IMU measurements propagate the state at high rate. GNSS, barometer, and magnetometer measurements then correct it via the error-state injection scheme [3]. The filter outputs a full state vector *and covariance* that are passed directly as the initial condition for the IPP filter.

Impact Point Prediction — Cartesian EKF

The IPP filter propagates a Cartesian state $\mathbf{x} = [\mathbf{p}^\top, \mathbf{v}^\top]^\top$ in NED coordinates forward until altitude reaches sea level:

$$\dot{\mathbf{p}} = \mathbf{v}, \qquad \dot{\mathbf{v}} = \mathbf{g} - K_D\,\mathbf{v}_a + \mathbf{R}(\hat{\mathbf{q}})\,\mathbf{T}_b, \qquad K_D = \frac{\rho\, C_D\, A\, \|\mathbf{v}_a\|}{2m}$$

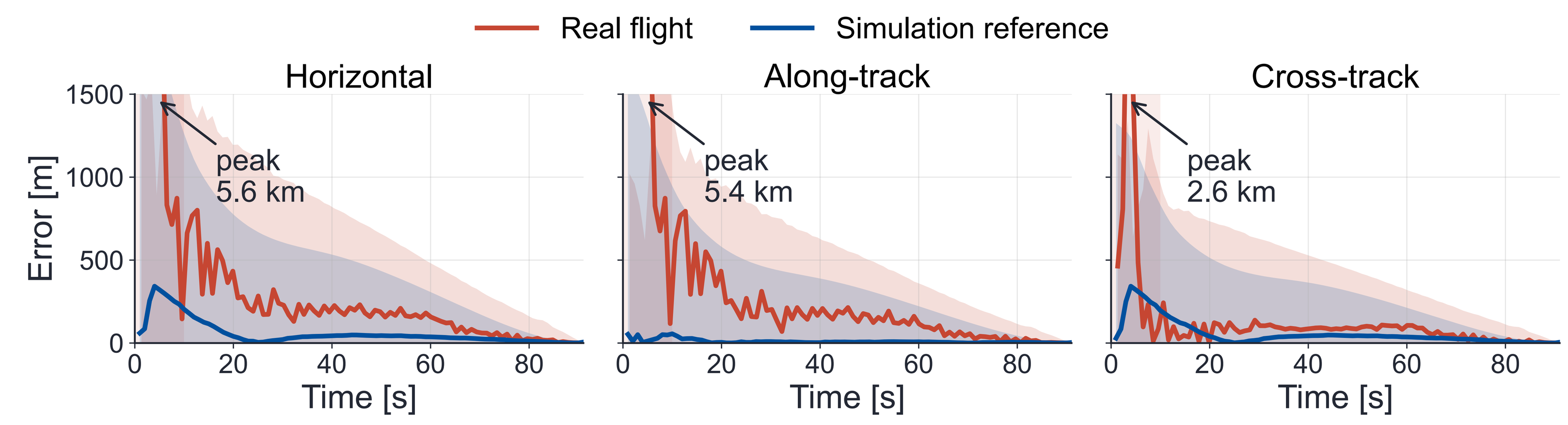
$\mathbf{v}_a$ airspeed vector, ρ air density, C_D drag coefficient, A reference area, m mass. $\mathbf{R}(\hat{\mathbf{q}})$ rotates the body-frame thrust $\mathbf{T}_b$ to NED using the ESKF attitude quaternion $\hat{\mathbf{q}}$.

The equations are propagated forward until the predicted altitude reaches sea level, yielding a predicted impact point and propagated uncertainty ellipse updated in real time.

Component	Key design choice
ESKF	Fuses GNSS, barometer, and magnetometer
IPP	Forward propagation of position and velocity using EKF equations
Interface	IPP re-initialised from ESKF state and covariance at each GNSS update

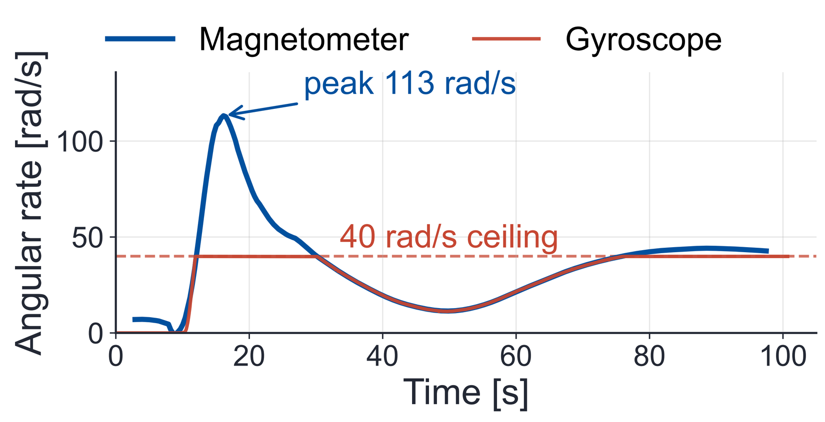
Results: Real-Flight Validation

Andøya student-rocket flight (Mongoose 98), full ~ 80 s ballistic descent to splashdown. The Cartesian IPP filter is run on real onboard sensor data alongside a simulation reference with clean sensor inputs as a baseline.



During the burn phase ($t \lesssim 10$ s) the prediction is corrupted by a stack of upstream effects, including gyroscope and accelerometer saturation under high spin, vibration, GNSS error, and thrust/mass mismodelling. After burn-out the real-flight error drops sharply and decays through to impact, settling above the simulation reference baseline. Errors are reported against the GNSS reference rather than absolute ground truth.

Challenges & Outlook



Spin saturation. The gyroscope saturates at ± 40 rad/s through burn. The magnetometer recovers the true ≈ 113 rad/s peak. Mitigations include a higher-range IMU, IMU placement closer to the spin axis, and a lower-spin rocket.

- **Wind and drag uncertainty.** Real-time wind estimation and in-flight drag identification remain open problems for operational use.
- **Known mass and thrust.** The IPP model treats vehicle mass and thrust as deterministic inputs. Propellant burn-off and thrust variations introduce unmodelled uncertainty that limits pre-burnout accuracy. In-flight parameter estimation would address this.
- **Scaling to orbital recovery.** The ESKF and Cartesian architecture requires only higher-grade inertial sensors and a longer propagation horizon to support autonomous ship pre-positioning before first-stage splashdown.

References

[1] K. Gryte et al. *Recovery of Rocket Payloads and First Stages Using Unmanned Vehicles.* IEEE Aerospace Conference, 2025.

[2] A. Meta. *Impact Point Prediction of High-Power Rockets and Sounding Rockets.* M.Sc. Thesis, NTNU, 2025.

[3] E. K. Jensen. *Rocket science: Catch space rockets using unmanned vehicles.* M.Sc. Thesis, NTNU, 2026.